

Quantum Supremacy in Finance — Marketing or Reality?

An Honest Accounting for 2026

Nyquist Research

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Abstract

Every major bank now has a quantum computing team. Not one financial workload runs in production on a quantum computer. A line-by-line audit of the four claims — derivative pricing, portfolio optimization, quantum machine learning, broken encryption — the honest timelines for each, and the one quantum task that already deserves engineering budget, governance, and execution today.

Nyquist Research 06 · Published 13 May 2026 · Reading time: 18 min · Topic: Quantum · Cryptography · Risk Infrastructure · Anchored against ~150 physical qubits · ~7,500 logical needed · NIST FIPS 203/204/205 · Five claims · current state · honest horizon · 1 actionable now · 4 deferred.

The only quantum task that deserves budget in 2026 is at the top of this list.

Each row is a claim a vendor has made on a slide in the last twelve months. Each row is evaluated against the same scale: theoretical strength of the result, current hardware reality, and the engineering horizon before a financial user should expect routine workload migration.

Layer	Claim / area	Description	Horizon
Now · Crypto	Post-quantum cryptography migration	NIST finalized ML-KEM, ML-DSA, SLH-DSA in August 2024. Harvest-now-decrypt-later is already happening. The only quantum work that belongs in a 2026 budget.	Immediate
L1 · Encryption	Shor's algorithm vs RSA-2048 / ECC	2025 Gidney estimate: under one million noisy qubits, under one week. Not imminent, but distance is shrinking faster than the comfort zone allows.	5–10 yr · serious threat

Layer	Claim / area	Description	Horizon
L2 · Pricing	Quantum amplitude estimation for derivative Monte Carlo	Strongest theory of the four claims. Resource estimate: ~7,500 logical qubits before classical GPU Monte Carlo loses on wall-clock.	8–12 yr · narrow use
L3 · Optimization	QAOA + annealing for constrained discrete portfolio problems	Classical Gurobi / CPLEX still wins on benchmark families that matter. Possibly narrow value in cardinality-constrained niches.	3–5 yr · niche · maybe
L4 · ML / alpha	Quantum machine learning for signal generation	No reproducible advantage on real financial datasets vs. strong classical baselines. Sells decks. Does not produce Sharpe.	Unclear · possibly never
L5 · Latency	Low-latency execution and market-data plumbing	FPGA at room temperature next to the exchange wins this contest permanently. No roadmap changes that.	Never · structural

One row is operational today. The others are research programs, scoring optionality, or category errors.

crypto-agility · not quantum-native · the right posture for 2026

Every major bank now seems to have a quantum computing team. IBM, Google, IonQ, D-Wave, Microsoft, and a long tail of vendors publish regular announcements suggesting that finance is one breakthrough away from a quantum rewrite. Goldman Sachs has published quantum pricing research. JPMorgan has built an internal quantum research capability. Consortium memberships and vendor partnerships have become standard signaling devices for innovation budgets.

And yet: **not a single production trading strategy runs on a quantum computer today**. Not one. No market-making engine, no intraday risk stack, no XVA platform, no limit-order-book model, no factor timing pipeline, no high-frequency execution framework. That gap between narrative and reality is not an embarrassment for the field; it is the central fact that any honest discussion has to start from.

The mistake in most quantum-finance commentary is that it is directionally wrong in two different ways at once. It is **overhyped** about the near-term use cases that dominate press releases — option pricing, alpha generation, and generic “portfolio optimization.” But it is also **underappreciated** where the first economically meaningful consequences may appear: crypto-agility, post-quantum migration, specific constrained optimization niches, and the institutional option value of building technical readiness before the hardware becomes useful.

That distinction matters. If the wrong claims dominate the conversation, decision-makers either allocate capital to science projects with no realistic deployment path, or dismiss the field entirely and miss the one area where action is already warranted. The right posture in 2026 is neither evangelism nor cynicism. It is *technical discrimination*.

The right posture is neither evangelism nor cynicism. It is technical discrimination — claim by claim, hardware threshold by hardware threshold. *The thesis of this paper.*

1 What “quantum supremacy” means — and what it does not

The most abused term in the field remains “quantum supremacy.” Google’s 2019 Sycamore result referred to a random circuit sampling task that the team said was completed in around 200 seconds, with a classical supercomputer estimate measured in thousands of years. That result was scientifically important because it established a milestone in controllable quantum hardware. But it was never an economically useful computation. The task was intentionally designed to be difficult for classical simulation and irrelevant to business problems.

That is why practitioners should mostly stop using the term unless the topic is hardware history. The term that actually matters for finance is **quantum advantage**: solving a practically useful problem faster, cheaper, or more accurately than the best classical alternative under realistic constraints. On that metric, as of 2026, no financial use case has demonstrated a defensible quantum edge in production.

The distinction is not semantic. A benchmark can prove that a machine behaves in a fundamentally quantum way. It does not prove that the machine is commercially useful. In finance, usefulness has a very specific meaning: lower total cost of computation for a required accuracy target, acceptable latency, integration into classical data systems, reproducibility, operational stability, and a workflow that survives model validation, audit, cyber review, and production support.

Three paradigms matter for this discussion:

- **Gate-based quantum computing** — IBM, Google, IonQ, Quantinuum, Rigetti. This is the universal model. In principle it can run algorithms such as Quantum Amplitude Estimation (QAE), Shor’s algorithm, HHL-type linear-system methods, QAOA, and various quantum machine-learning routines. In practice, current devices remain noisy and limited.
- **Quantum annealing** — mainly D-Wave. This is not universal quantum computing; it is a specialized optimization architecture for problems that can be formulated in Ising / QUBO form. The right comparison is not “general quantum vs. classical computing,” but “specific constrained optimization heuristics vs. classical heuristics and commercial solvers.”
- **Photonic / topological ambitions** — PsiQuantum, Xanadu, and Microsoft’s topological path. These approaches are strategically important because they target fault tolerance differently, but they are not yet a source of near-term production finance workloads.

The hardware bottleneck remains decisive. IBM’s public roadmap points toward increasingly modular systems and toward fault-tolerant milestones later in the decade, with 2029 highlighted as a target for fault-tolerant quantum computing. Google’s Willow processor, announced in late 2024, featured 105 qubits and a new benchmark result linked to error correction progress, but the result still does not imply readiness for large-scale commercial algorithms. On the current generation of hardware, the industry is still measuring progress in physical-qubit quality, error suppression, and modular scaling — not in deployable logical-qubit abundance.

That last phrase is the important one: **logical qubits**. A finance-relevant quantum computer is not “a machine with hundreds or thousands of physical qubits.” It is a machine with enough error-corrected logical qubits to run deep circuits reliably. Under conventional fault-tolerance assumptions, one logical qubit may require hundreds to thousands of physical qubits depending on architecture and target error rate. That is why seemingly impressive physical-qubit counts can still correspond to effectively zero useful logical compute for realistic financial workloads.

A reasonable 2026 base case is therefore blunt: current hardware is scientifically meaningful, commercially premature, and still several engineering layers away from the threshold at which financial users should expect routine workload migration.

2 The finance claims — audited one by one

2.1 Claim 1 · “Quantum computers will price derivatives faster.”

This is the strongest claim **in theory** and one of the weakest **in deployment timing**. The mathematical basis is well known: Quantum Amplitude Estimation can reduce sample complexity from classical Monte Carlo’s roughly $O(1/\varepsilon^2)$ to $O(1/\varepsilon)$, giving a quadratic improvement in accuracy scaling. For computational finance people, that is not marketing fluff. If realizable, it is a very serious asymptotic result.

This is why the Goldman Sachs / IBM / ETH Zürich line of work attracted attention. The paper *Option Pricing using Quantum Computers* formalized how to represent option-pricing problems in a quantum framework and showed where a quantum advantage might arise. The conceptual point remains valid today: if one must estimate expectations under complex path-dependent distributions with tight error tolerances, then a quadratic advantage over Monte Carlo is not trivial; it is potentially transformative.

But asymptotic speedup is not the same as wall-clock superiority. In practice, classical Monte Carlo in finance is not standing still. It runs on multi-core CPUs, GPUs, variance-reduction frameworks, adjoint methods, quasi-random sequences, surrogate models, and workflow-specific approximations. A vanilla option is priced in milliseconds on modern GPU infrastructure. Even many exotics are dominated by model setup, calibration, market-data handling, and control-variate engineering rather than raw sampling cost.

The most important reality check comes from the same research line that motivates the excitement. Stamatoopoulos et al. estimated that meaningful quantum advantage for option pricing could require on the order of **7,500 logical qubits**. That is not a minor engineering gap from where hardware sits today. It is a different era of computing.

The practical implication is uncomfortable but clear. Near-term vendor claims that imply a quantum edge in front-office derivative pricing are mostly category errors. The theoretical result is real. The near-term commercial story is not. The first places where QAE-like methods might matter are not daily listed-vanilla workflows, but computationally severe tail cases: high-dimensional exposures, nested simulation settings, or specialized exotic structures where classical approximation error and compute budgets remain genuinely painful.

*Honest verdict — strong theory, real long-run promise, **no present trading advantage**, and probably an 8–12 year horizon for narrow production relevance even under favorable hardware progress assumptions.*

2.2 Claim 2 · “Quantum optimization will revolutionize portfolio construction.”

This is the most common sales pitch because it sounds intuitive. Finance has many optimization problems. Quantum devices are often introduced as optimization machines. Therefore, portfolio construction should be a natural killer app. The argument is rhetorically elegant and technically incomplete.

First, one has to separate **continuous convex optimization** from **combinatorial constrained optimization**. Classical mean-variance optimization, risk-budgeting, and many institutionally relevant allocation problems are already solved extremely well by classical convex optimizers. For these tasks, “quantum optimization” is usually solving the wrong problem. If a portfolio problem is convex, well-conditioned, and moderate in size, commercial solvers and classical numerical libraries are already excellent.

Where the conversation becomes more serious is the QUBO / Ising class: cardinality constraints, turnover penalties, lot sizes, sector caps, tracking-error objectives with discrete selection, index replication under sparse holdings, scenario-tree design, and various mixed-integer formulations. These are harder. They are also the class that D-Wave and QAOA advocates target most directly.

The evidence so far is mixed at best. Portfolio-optimization studies on D-Wave annealers have shown feasibility and occasionally competitive solutions, but not a robust superiority over classical solvers. More broadly, comparative studies published in 2024 found that classical methods still outperform quantum annealing and hybrid pipelines on most relevant benchmark families, especially once solution quality,

repeatability, and end-to-end runtime are evaluated honestly rather than selectively.

The gate-based story is even more constrained. QAOA is elegant and influential as a research program, but the circuit depths needed for nontrivial financial instances often exceed coherence windows on current devices. Parameter training itself is also nontrivial. Once one includes embedding overhead, repeated measurements, and classical outer-loop optimization, the advantage story becomes much harder to defend.

Still, it would be wrong to dismiss the area entirely. A 2024 JPMorgan Chase / AWS collaboration on decomposition pipelines for large-scale portfolio optimization is an example of the right kind of institutional research. It does not pretend that today’s hardware wins outright. Instead, it asks a more credible question: can large problems be restructured so that a future quantum subroutine solves the most combinatorially expensive component while the classical stack does everything else? That is a genuine architecture question, not a press release.

*Honest verdict — classical solvers still win decisively for mainstream portfolio construction; annealing and hybrid quantum-classical methods may find **narrow value** in highly constrained discrete problems over a 3–5 year horizon; universal gate-based QAOA for production portfolio construction remains further out.*

2.3 Claim 3 · “Quantum machine learning will improve alpha generation.”

This is where skepticism should be highest. Quantum machine learning has generated an enormous academic literature and an even larger marketing halo. The typical narrative is that quantum feature maps, quantum kernels, or variational circuits provide richer representation spaces that should unlock superior predictive structure in noisy financial data.

That is a very strong claim, and the burden of proof should be correspondingly high. In finance, alpha-generation pipelines already sit on top of mature classical ecosystems: gradient boosting, random forests, linear and nonlinear factor models, transformers, graph models, sequence models, Bayesian state-space systems, and large-scale feature engineering on GPU and distributed clusters. Any credible quantum-ML claim must beat not a toy baseline, but this stack.

So far, it has not. There is no established body of evidence showing a reproducible outperformance of quantum ML over strong classical baselines on real financial datasets. The basic reasons are structural. Financial data is noisy, weak-signal, regime-dependent, nonstationary, and relatively small in true signal-to-noise once leakage is controlled. NISQ hardware adds noise exactly where one needs stable estimation. The input bottleneck is also severe: classical data must be encoded into quantum states, and that loading cost can erase theoretical gains.

Even if one grants that some quantum kernels generate classically hard feature spaces, that does not automatically convert into better out-of-sample Sharpe ratios. Finance is not a contest in abstract expressivity; it is a contest in robust, transaction-cost-aware, adaptive prediction under strict controls. The absence of convincing empirical advantage is not an accident. It is a sign that the problem formulation has been ahead of the hardware and perhaps ahead of the economics.

*Honest verdict — quantum ML is a live academic field and an **extremely weak commercial claim** for 2026. Anyone selling near-term quantum alpha should be asked for benchmark design, transaction-cost assumptions, out-of-sample stability, model-risk controls, and classical baseline strength. Most claims do not survive that questioning.*

2.4 Claim 4 · “Quantum will break financial encryption — this is an existential risk.”

This claim is mathematically the most robust and operationally the most misunderstood. Shor’s algorithm can factor integers and solve discrete logarithms in polynomial time on a sufficiently powerful fault-tolerant quantum computer. That means RSA and elliptic-curve cryptography are, in principle, breakable by future quantum machines. This is not speculative; it is one of the foundational results of quantum computing.

The old comfort was timeline distance. Gidney and Ekerå’s 2021 estimate suggested that factoring RSA-2048 in around 8 hours would require roughly 20 million noisy qubits under plausible assumptions,

corresponding to around 4,000 logical qubits in broad shorthand. That was already serious, but it made the risk feel comfortably deferred.

Then the estimates started tightening. In 2025, Craig Gidney published updated work suggesting that RSA-2048 could be factored with **fewer than one million noisy qubits in under a week** under revised algorithmic assumptions. That does not mean the threat is imminent in the next quarter. It does mean the distance between “theoretically breakable” and “engineering-plausible within a strategic horizon” is shrinking.

More importantly, the relevant risk is not only the day a quantum computer can decrypt live traffic. The relevant risk is **harvest now, decrypt later**. Adversaries can capture encrypted data today and store it for future decryption once hardware matures. That is why institutions with long confidentiality horizons are already exposed. Regulatory communications, client records, internal architecture documents, credentials, archival messages, proprietary strategies, and research archives may retain value well into the quantum-relevant window.

This is no longer just a cybersecurity talking point. NIST finalized the first three post-quantum cryptography standards — **FIPS 203, FIPS 204, and FIPS 205** — in August 2024, standardizing ML-KEM, ML-DSA, and SLH-DSA families for quantum-resistant use. The Federal Reserve published a 2026 paper explicitly discussing the harvest-now-decrypt-later problem and its implications for financial systems. That is the signal practitioners should notice. Unlike most quantum-finance narratives, this one has already crossed from research curiosity into institutional planning.

*Honest verdict — live quantum decryption of financial infrastructure is not here yet, but **post-quantum migration is the one quantum task that deserves budget, governance, and execution right now.***

3 A practical verdict table

Five claims, evaluated against the same scale: theoretical basis, current reality, and the engineering horizon before each one becomes a credible budget item.

Claim	Theoretical basis	Current reality	Horizon
Derivative pricing via QAE	Strong; quadratic sample-complexity improvement	Hardware gap remains enormous; estimated logical-qubit requirements far above current capability	8–12 yr · narrow
Portfolio optimization via QAOA / annealing	Moderate; better story for discrete constrained problems than for standard MVO	Classical solvers still dominate mainstream production workloads	3–5 yr · niche · maybe
Quantum ML for alpha	Weak commercial evidence	No demonstrated, reproducible advantage on real financial datasets	Unclear · possibly never
Breaking encryption with Shor	Strong and mathematically settled	Not operationally feasible yet, but resource estimates are improving	5–10 yr · serious threat
Harvest now, decrypt later	Strong strategic risk already recognized by institutions	Happening now wherever long-lived sensitive data is collected and stored	Immediate action

4 What practitioners should actually do now

Three tiers. They are not equally urgent. They should not be equally resourced.

4.1 Tier 1 · 0–12 months · Immediate

Audit cryptographic exposure. Build crypto-agility before anything else.

The first task is not to build a quantum desk. It is to audit cryptographic exposure. Many trading firms, fintech platforms, prime-broker interfaces, and internal APIs still rely on RSA or ECC in authentication flows, transport layers, code-signing chains, VPNs, hardware modules, key exchange, and vendor connectivity. **Without a migration inventory, there is no quantum-readiness strategy — only slogans.** The right mindset here is not “replace everything tomorrow.” It is “stop being structurally unprepared.” For many institutions, governance and architecture debt — not math — will be the hardest part of migration.

The serious immediate program in Tier 1 includes:

- cryptographic asset inventory across internal systems, client-facing APIs, key management, and vendor channels;
- identification of datasets with long confidentiality tails, especially strategy IP, regulator-facing material, client communications, and archived research;
- crypto-agility planning so that algorithms can be swapped without rewriting the full platform;
- limited testing of NIST-standardized post-quantum schemes in low-criticality environments to build operational experience.

4.2 Tier 2 · 1–3 years · Selective

Selective experimentation, not broad deployment.

Monitor quantum optimization benchmarks with real skepticism, not vendor case studies as evidence. The most credible focus areas are discrete constrained optimization problems where classical heuristics are imperfect and the objective structure maps naturally into annealing or hybrid workflows. Evaluate hybrid quantum-classical orchestration models. **The point is not to force quantum into the stack. The point is to design research and optimization layers so that a future quantum subroutine can be inserted without rewriting the surrounding infrastructure.** Architectural optionality matters more than near-term performance theater.

4.3 Tier 3 · 3–10 years · Watch & wait

Track milestones, not press releases.

For pricing, the relevant milestone is not another paper showing asymptotic speedup; it is progress toward **stable logical qubits at the scale needed for deep amplitude-estimation circuits**. For optimization, the relevant milestone is repeatable advantage on economically meaningful constrained instances against hard classical baselines. For quantum ML, the relevant milestone is simply the first robust empirical result that survives strong classical benchmarking. Reserve strategic attention for these. Resist the temptation to front-run the hardware with product commitments. In quantum finance, premature integration is more likely to create technical debt than edge.

5 Who is actually ahead — and who is mostly signaling

Among financial institutions, the genuinely serious actors are not hard to identify. Goldman Sachs has consistently published substantive work on quantum algorithms for pricing and risk-related computations,

including the well-known option-pricing collaboration. JPMorgan Chase has maintained a visible and technically credible research effort, including optimization work with Quantinuum and portfolio decomposition research with AWS. These efforts matter because they build internal expertise, benchmark discipline, and institutional memory.

But the honest assessment is that even the leaders are not “deploying quantum finance” in the popular sense. They are buying **optionality**. They are placing informed, relatively small bets on talent, partnerships, experimentation, and IP so that they are not structurally late if and when the hardware crosses a useful threshold. *That is different from production adoption.*

At the other end of the spectrum sit the overhyped announcements: generic memoranda of understanding, cloud-access partnerships framed as strategic transformation, and token innovation narratives with little evidence of differentiated research. Many bank quantum teams remain small — often a handful of researchers — and are still doing what they should be doing in 2026: exploration, not deployment.

The distinction matters for executives. A press release saying a bank “is working with quantum computing” may mean anything from deep algorithmic research to a vendor access agreement and one internship cohort. **Those are not equivalent signals.**

6 The contrarian view — where quantum may never win

There is another mistake in quantum commentary: even serious analysts often assume that once hardware matures, quantum will gradually absorb a wide range of financial workloads. That may be false. There are large parts of the financial technology stack where quantum may never be the right tool.

Low-latency trading is the clearest case. Quantum computers run in cryogenic environments and operate with gate times that make them fundamentally irrelevant for sub-millisecond execution problems. An FPGA sitting next to an exchange gateway at room temperature wins this contest permanently. No plausible roadmap changes that economic fact.

Data-intensive systems are another likely non-domain. Financial computation is often bottlenecked not by arithmetic alone, but by data movement, normalization, cleaning, reconciliation, and state management. Quantum algorithms usually assume that data is already encoded into a quantum state at acceptable cost. In real systems, that assumption is often heroic. The input/output bottleneck can erase the benefit before the algorithm starts.

General-purpose platform workloads — order management, market-data plumbing, audit trails, surveillance, backtesting infrastructure, risk aggregation, and most of the software surface area of a modern trading platform — will remain classical. Even in a mature quantum future, the most realistic architecture is a classical platform with rare calls to specialized quantum co-processors for well-defined bottlenecks.

There is also a real **quantum winter** scenario. If large-scale error correction proves materially harder than current roadmaps imply, then useful fault-tolerant quantum computing could be delayed by another decade. Microsoft’s 2025 Majorana announcement illustrates the point: the field remains capable of spectacular claims that are then subjected to equally spectacular scrutiny. For decision-makers, this means *timeline humility is not optional. It is rational governance.*

7 Where Nyquist fits

Nyquist’s position is straightforward: build classical infrastructure that is **quantum-aware**, not *quantum-native*. That means treating post-quantum cryptography as a present architecture problem rather than a future marketing theme. It means keeping optimization and analytics layers modular enough that, if a genuine advantage emerges in annealing or hybrid quantum routines, workloads can be routed selectively without rebuilding the platform. And it means refusing to anchor product strategy to vendor roadmaps that remain highly uncertain.

That posture may sound conservative. It is actually the most aggressive version of realism. The wrong way to prepare for quantum is to pretend it is already here. The right way is to ensure that today’s

infrastructure does not become tomorrow’s security liability — and that tomorrow’s compute primitives can be integrated without institutional replatforming.

The wrong way to prepare for quantum is to pretend it is already here. The right way is to ensure today’s infrastructure does not become tomorrow’s security liability.

Nyquist’s posture · 2026 onward

8 What the industry should stop saying

A more honest quantum-finance vocabulary would retire several phrases immediately:

- **“Quantum will soon replace Monte Carlo”** — false in any operational sense relevant to 2026.
- **“Quantum optimization beats classical portfolio construction”** — false as a general statement and usually false even in narrowly defined benchmarks.
- **“Quantum ML will unlock alpha”** — unsupported by convincing empirical evidence.
- **“Quantum is too far away to matter”** — false in cyber and cryptographic planning.
- **“A bank quantum partnership implies strategic lead”** — often just signaling.

The better framing is narrower and stronger. Quantum computing is a long-duration technology with potentially profound implications for a small number of computational problems. Finance is one of the sectors where those implications may eventually matter because it already sits on Monte Carlo, optimization, linear algebra, cryptography, and large-scale simulation. But **the path from theoretical fit to commercial edge is bottlenecked by hardware, error correction, systems integration, and economics** — not by ambition.

9 Key takeaways

Seven sentences worth keeping.

1. **No quantum computer today solves a real financial problem faster than the best classical production stack.**
2. Quantum advantage in derivative pricing is *theoretically credible but hardware-constrained*, with meaningful relevance likely 8–12 years away.
3. Portfolio optimization is the most overmarketed near-term use case; any practical edge is more likely to appear in narrow discrete constrained problems than in standard allocation engines.
4. Quantum ML remains mostly an academic and branding story rather than a proven alpha technology.
5. **Post-quantum cryptography is the only quantum topic that deserves immediate operational attention**, because the harvest-now-decrypt-later threat exists before the day of live decryption.
6. The institutions that are “ahead” are not running quantum strategies; they are accumulating expertise and architectural optionality.
7. A realistic quantum strategy for a financial platform in 2026 is *modularity, crypto-agility, and disciplined monitoring* — not product theater.

10 Closing question

Is the firm’s quantum agenda still sitting in the “interesting but not urgent” bucket — or has post-quantum migration already been elevated to an engineering and risk-management program?

If the discussion is serious — especially around post-quantum cryptography, crypto-agile trading infrastructure, and modular compute architecture for future optimization workloads — the conversation is worth having now, **before the headlines and the real timelines converge.**

11 Inputs and assumptions

Sources and anchor points

Hardware references

IBM public roadmap and 2026 roadmap materials, including a stated path toward fault tolerance by 2029.

Google reference

Willow processor announcement and related reporting in late 2024; **105 qubits**, benchmark framed around error-correction progress rather than finance-ready utility.

Option-pricing estimate

Stamatopoulos et al., *Option Pricing using Quantum Computers*. The ~7,500-logical-qubit threshold is the load-bearing number for QAE on real financial workloads.

RSA-2048 resource estimates

Gidney & Ekerå (2021) and updated Gidney work in 2025 reducing noisy-qubit estimates materially — under one million noisy qubits, under one week.

Post-quantum standards

NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA) — finalized August 2024.

Federal Reserve note

2026 working paper explicitly discussing harvest-now-decrypt-later implications for the financial system.

Quantum advantage timeline

Based on public vendor roadmaps, recent research, and conservative interpretation of hardware and error-correction bottlenecks rather than vendor marketing claims.

Classical benchmarks

Commercial optimizers such as Gurobi / CPLEX for portfolio construction; GPU-accelerated Monte Carlo stacks for derivative pricing.

12 Constraints, limitations, and sensitivity

Where this analysis can be wrong, and the direction of its bias.

- Quantum-computing timelines remain highly unreliable; both excessive optimism and excessive pessimism have repeatedly been wrong. This paper leans toward the cautious end of the published range.
- “Quantum advantage” is problem-structure dependent. A method that is irrelevant at one scale may become interesting at another, especially for discrete combinatorial problems.
- Resource estimates for fault tolerance may improve with better codes, architectures, compilers, and algorithmic refinements; the 2025 update to factoring estimates is a reminder that the moving parts are nontrivial.

- This analysis focuses on the NISQ and early fault-tolerant era. Post-2035 possibilities remain genuinely uncertain and should not be overconfidently extrapolated from current constraints.
- Post-quantum migration complexity varies dramatically across institutions. Firms with legacy infrastructure, fragmented identity stacks, and long-lived archived data are materially more exposed than greenfield platforms.
- The article intentionally treats vendor claims skeptically and may underweight upside scenarios in which error correction, modular scaling, or alternative qubit architectures improve faster than current consensus expects.

Nyquist publishes Field Notes — write-ups from the team building the platform — when an industry conversation has drifted far enough from the underlying engineering that it is worth pulling back to the constraints. This paper is one of those.

The most useful response to it is not agreement. It is a sharper, more specific disagreement on any single claim — *with the math attached*.